

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6: CONDENSED-MATTER PHYSICS

TRINITY TERM 2019

Thursday, 13 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Define the terms *lattice* and *basis* as they have been used in this course. [2]

(b) Given a lattice of points $\mathbf{R}$ with primitive lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$, define the reciprocal lattice as the set of points $\mathbf{G}$ in reciprocal space such that

$$\exp(i \mathbf{G} \cdot \mathbf{R}) = 1$$

for all points $\mathbf{R}$ in the lattice. Give an expression for the primitive lattice vectors of this reciprocal lattice in terms of the vectors $\mathbf{a}_i$ and prove that the vectors you have given are indeed primitive reciprocal lattice vectors, that is, show that all $\mathbf{G}$ satisfying the equation above for all $\mathbf{R}$ are generated from your chosen primitive reciprocal lattice vectors. [7]

(c) Consider a function $V(\mathbf{x})$ which has the periodicity of the crystal lattice, meaning that $V(\mathbf{x}) = V(\mathbf{x} + \mathbf{R})$ for all lattice vectors $\mathbf{R}$. Show that the Fourier transform of this function

$$V_{\mathbf{q}} = \int d\mathbf{x} \exp(i \mathbf{q} \cdot \mathbf{x}) V(\mathbf{x})$$

is zero unless $\mathbf{q}$ is a reciprocal lattice vector. [3]

(d) The scattering potential felt by an electron as it propagates through the crystal is given by

$$V(\mathbf{x}) = \sum_j u_j(\mathbf{x} - \mathbf{x}_j)$$

where the sum over j is the sum over all the atoms in the entire system, $\mathbf{x}_j$ is the position of atom j , and u_j is a potential function that depends on the particular type of atom j . The transition rate for the electron to scatter from incoming wavevector $\mathbf{k}$ to outgoing wavevector $\mathbf{k}'$ is given by

$$\Gamma(\mathbf{k}', \mathbf{k}) \propto |\langle \mathbf{k}' | V(\mathbf{x}) | \mathbf{k} \rangle|^2.$$

Derive the following expression for the intensity I of an electron beam scattered by a reciprocal lattice vector $\mathbf{G} = \mathbf{k}' - \mathbf{k}$

$$I(\mathbf{G}) \propto |S(\mathbf{G})|^2$$

with

$$S(\mathbf{G}) = \sum_p b_p(\mathbf{G}) \exp(i \mathbf{G} \cdot \mathbf{r}_p)$$

where here the sum on p is a sum over the atoms in only a single unit cell, and $\mathbf{r}_p$ is the position of atom p within the unit cell. Give an expression for the function $b_p(\mathbf{G})$ in terms of the potential functions $u_j(\mathbf{x})$. Make sure to specify which sums are over the entire system and which sums are only within a single unit cell. [7]

(e) Crystalline silicon has a face-centred cubic lattice with two atoms per unit cell at positions $[0, 0, 0]$ and $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ in units of the lattice constant a . In a powder diffraction experiment on a sample of crystalline silicon, what are the magnitudes $|\mathbf{G}| = |\mathbf{k} - \mathbf{k}'|$ corresponding to the two lowest angle scattering rings observed? [6]

2. (a) State the assumptions of the Einstein model of atomic vibrations in a solid. Using the Einstein model, derive an expression for the heat capacity of a solid containing N identical atoms. [6]

(b) In the high temperature limit, the heat capacity can be expanded as

$$C = A_0 + A_1/T + A_2/T^2 + \dots$$

where T is temperature. Using the series expansion

$$\frac{1}{e^x - 1} = \frac{1}{x} - \frac{1}{2} + \frac{x}{12} + \dots$$

for small x , or otherwise, give values for the coefficients A_0 , A_1 and A_2 . [3]

[Hint: it is easiest to expand the expression for energy.]

(c) At high temperature the heat capacity of a solid sample of argon is given by

$$C = (25 - 7500/T^2 + \dots) \text{ J K}^{-1},$$

where T is measured in Kelvin. Compare this data to the above expression for the Einstein heat capacity at high temperature to determine the Einstein temperature T_E of argon. [3]

(d) Given that argon has atomic weight of 40, what is the mass of this sample? Given that argon forms a face-centred cubic lattice with one atom per unit cell and the volume of the sample is 21 cm^3 , find the nearest-neighbour distance between argon atoms. [5]

(e) Using the Einstein picture of argon, derive an expression applicable for $T \gg T_E$ for $\langle x^2 \rangle$, the mean squared fluctuation of the x -coordinate of an atom, as a function of T , T_E , and the mass of the argon atom. [5]

(f) Frederick Lindemann conjectured that a face-centred cubic crystal will melt when $\sqrt{\langle x^2 \rangle} = 0.05 d$, with d the atomic nearest neighbour spacing. Using this conjecture, estimate the melting temperature of argon. [3]

3. A two-dimensional semiconductor may be constructed using a so-called quantum well geometry—a very thin layer of semiconductor sandwiched between two very good insulators. You may assume that electrons or holes in this layer can move in two dimensions only.

(a) Assuming the chemical potential μ is near the middle of the bandgap so that Boltzmann statistics applies, the density n of electrons near the bottom of the two-dimensional conduction band is given by

$$n = C (m_e^*)^a T^b e^{-\beta(E_c - \mu)}$$

where E_c is the energy of the bottom of the conduction band, m_e^* is the effective mass of electrons at the bottom of the conduction band, T is temperature, and $\beta = 1/(k_B T)$. Derive values for the constant C and the exponents a and b . [7]

(b) Give a similar expression for the density p of holes near the top of the two-dimensional valence band. Give an expression for the product np . [3]

(c) A two-dimensional semiconductor quantum well is made with InAs as the quantum well layer. Determine the density of electrons in the conduction band at $T = 300$ K (assuming the system is undoped). Some useful parameters for this material are listed below. Suppose the two-dimensional system is now doped with $n_d = 10^9 \text{ cm}^{-2}$ donors, estimate the temperature at which the sample changes from intrinsic to extrinsic behavior. [5]

(d) If a donor impurity is added at position $\mathbf{r}_0$ to the two-dimensional layer the donated electron may bind to the donor nucleus. The form of the bound electron wavefunction is

$$\psi(\mathbf{r}) = A e^{-b|\mathbf{r} - \mathbf{r}_0|}$$

where $\mathbf{r}$ is the position in the two-dimensional plane and $\mathbf{r}_0$ is the position of the donor nucleus. Using the Schrödinger equation in polar coordinates, or otherwise, determine the constant b and the binding energy. [5]

(e) At 200 K, the two-dimensional conductivity σ of the doped sample is given by $\sigma = 0.01 \Omega^{-1}$. Make a rough labelled sketch of $\log \sigma$ as a function of $1/T$ for temperatures between $T = 30$ and 600 K. You may assume that the mobilities of electrons and holes are the same and are independent of temperature. [5]

[Useful parameters for two-dimensional InAs:

Relative permittivity: $\epsilon_r = 15$

Band gap: $E_g = 0.36 \text{ eV}$

Electron effective mass: $m_e^* = 0.026 m_e$

Hole effective mass: $m_h^* = 0.41 m_e$

where m_e is the bare electron mass.]

4. (a) Define the terms *paramagnet*, *diamagnet*, and *ferromagnet*. [3]

(b) What kind of magnetic properties would you expect for helium? Derive an expression for the magnetic susceptibility of ^4He at low temperature in terms of the helium density, the mean squared radius of electrons in the helium atom, and fundamental constants. [6]

(c) The isotope ^3He is electronically identical to ^4He but has atomic weight of 3 rather than 4. The nucleus of ^3He has spin $1/2$ whereas the nucleus of ^4He has no spin. The magnetic moment of the nucleus of ^3He is given by $\mu_{^3\text{He}} = -2.13 \mu_N$ where μ_N is the nuclear magneton. Assuming the nuclei are independent, give an expression for the nuclear magnetic susceptibility of ^3He as a function of temperature and density. [6]

(d) At approximately what temperature would you expect the total magnetic susceptibility to change sign? [3]

(e) At $T < 4\text{ K}$ and pressures near 100 atmospheres, ^3He forms a body-centred cubic crystal. Careful measurement at very low temperature shows that the ^3He nuclei interact with each other via a Hamiltonian of the form

$$H = -\frac{1}{2} \sum_{i,j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$

where $\mathbf{S}_i$ are nuclear spins. Here we will assume that $J_{ij}/k_B = 1\text{ mK}$ if i and j are nearest neighbours, $J_{ij}/k_B = -0.5\text{ mK}$ if i and j are next-nearest neighbours, and $J_{ij} = 0$ otherwise. Using mean field theory, calculate the critical temperature for ferromagnetism. [7]